

BrainBolt - Answer Key

Topic: Java (15 Questions)

Q1. Which keyword is used to inherit a class in Java?

- A. extends <- correct
- B. implements
- C. inherits
- D. super

Answer: A

Why: The 'extends' keyword is used for class inheritance; 'implements' is for interfaces.

Q2. Which is NOT a primitive data type in Java?

- A. int
- B. boolean
- C. String <- correct
- D. char

Answer: C

Why: String is a class (java.lang.String), not a primitive. The primitives are byte, short, int, long, float, double, char, boolean.

Q3. What is the entry point of every Java program?

- A. start()
- B. run()
- C. main() <- correct
- D. init()

Answer: C

Why: JVM calls public static void main(String[] args) to start execution.

Q4. Which OOP concept is achieved using interfaces?

- A. Encapsulation
- B. Abstraction <- correct
- C. Inheritance
- D. Polymorphism

Answer: B

Why: Interfaces hide implementation details and expose only behaviour — that's abstraction.

Q5. Which keyword prevents method overriding?

- A. static

- B. final <- correct
- C. private
- D. abstract

Answer: B

Why: A method declared 'final' cannot be overridden by subclasses.

Q6. Which collection class allows duplicate values and maintains insertion order?

- A. HashSet
- B. TreeSet
- C. ArrayList <- correct
- D. HashMap

Answer: C

Why: ArrayList is an ordered list that allows duplicates.

Q7. What does JVM stand for?

- A. Java Virtual Machine <- correct
- B. Java Verified Method
- C. Joint Virtual Machine
- D. Java Variable Memory

Answer: A

Why: JVM = Java Virtual Machine — it executes Java bytecode.

Q8. Which operator is used for comparing two values for equality?

- A. =
- B. == <- correct
- C. ===
- D. equals

Answer: B

Why: == compares primitive values (and references for objects). .equals() compares object contents.

Q9. Which exception is thrown when dividing an integer by zero?

- A. NullPointerException
- B. ArithmeticException <- correct
- C. NumberFormatException
- D. IOException

Answer: B

Why: Integer division by zero throws java.lang.ArithmeticException.

Q10. Which keyword is used to refer to the current object?

- A. self
- B. current
- C. this <- correct
- D. me

Answer: C

Why: 'this' refers to the current instance inside an instance method or constructor.

Q11. What is the size of an int in Java?

- A. 2 bytes
- B. 4 bytes <- correct
- C. 8 bytes
- D. Depends on platform

Answer: B

Why: int is always 32 bits (4 bytes) in Java regardless of platform.

Q12. Which keyword is used to create an object in Java?

- A. new <- correct
- B. create
- C. object
- D. make

Answer: A

Why: The 'new' keyword allocates memory and invokes the constructor: e.g., new Car().

Q13. Which package contains String class?

- A. java.util
- B. java.io
- C. java.lang <- correct
- D. java.net

Answer: C

Why: String is part of java.lang, which is imported automatically by every Java program.

Q14. Which feature of OOP shows the same method behaving differently?

- A. Inheritance
- B. Polymorphism <- correct
- C. Encapsulation
- D. Abstraction

Answer: B

Why: Polymorphism (overloading & overriding) allows the same method name to behave differently in different contexts.

Q15. What is the default value of a boolean variable in Java?

- A. true
- B. false <- correct
- C. 0
- D. null

Answer: B

Why: Instance boolean variables default to false.